

EXERCISE - 08

4.

```
SELECT
    ROUND(MAX(salary)) AS Maximum,
    ROUND(MIN(salary)) AS Minimum,
    ROUND(SUM(salary)) AS Sum,
    ROUND(AVG(salary)) AS Average
FROM employees;
```

Output:

MAXIMUM	MINIMUM	SUM	AVERAGE
24000	2500	123800	8253

5.

```
SELECT job_id,
    MIN(salary) AS Minimum,
    MAX(salary) AS Maximum,
    SUM(salary) AS Sum,
    ROUND(AVG(salary)) AS Average
FROM employees
GROUP BY job_id;
```

Output:

JOB_ID	MINIMUM	MAXIMUM	SUM	AVERAGE
AD_VP	17000	17000	17000	17000
HR_REP	4500	4500	4500	4500
ST_CLERK	2500	3500	6000	3000
FIN_MGR	7000	7100	14100	7050
IT_PROG	5000	9000	25200	6300

6.

```
SELECT job_id, COUNT(*) AS Number_of_People
FROM employees
WHERE job_id = 'IT_PROG'
GROUP BY job_id;
Output:
```

JOB_ID	NUMBER_OF_PEOPLE
IT_PROG	4

7.
SELECT COUNT(DISTINCT manager_id) AS Number_of_Managers
FROM employees
WHERE manager_id IS NOT NULL;

Output:

NUMBER_OF_MANAGERS
7

8.
SELECT (MAX(salary) - MIN(salary)) AS Difference
FROM employees;

Output:

DIFFERENCE
21500

9.
SELECT manager_id,
MIN(salary) AS Lowest_Salary
FROM employees
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING MIN(salary) > 6000
ORDER BY Lowest_Salary DESC;

Output:

MANAGER_ID	LOWEST_SALARY
103	9000
102	8000
106	7000

10.

```
SELECT
    COUNT(*) AS Total_Employees,
    COUNT(CASE WHEN EXTRACT(YEAR FROM hire_date) = 1995 THEN 1
END) AS Hired_1995,
    COUNT(CASE WHEN EXTRACT(YEAR FROM hire_date) = 1996 THEN 1
END) AS Hired_1996,
    COUNT(CASE WHEN EXTRACT(YEAR FROM hire_date) = 1997 THEN 1
END) AS Hired_1997,
    COUNT(CASE WHEN EXTRACT(YEAR FROM hire_date) = 1998 THEN 1
END) AS Hired_1998
FROM employees;
```

Output:

TOTAL_EMPLOYEES	HIRED_1995	HIRED_1996	HIRED_1997	HIRED_1998
15	0	0	0	5

11.

```
SELECT job_id,
    SUM(CASE WHEN department_id = 20 THEN salary END) AS Dept20,
    SUM(CASE WHEN department_id = 50 THEN salary END) AS Dept50,
    SUM(CASE WHEN department_id = 80 THEN salary END) AS Dept80,
    SUM(CASE WHEN department_id = 90 THEN salary END) AS Dept90,
    SUM(salary) AS Total_Salary
FROM employees
WHERE department_id IN (20, 50, 80, 90)
GROUP BY job_id;
```

Output:

JOB_ID	DEPT20	DEPT50	DEPT80	DEPT90	TOTAL_SALARY
AD_VP	-	-	-	17000	17000
HR_REP	4500	-	-	-	4500
ST_CLERK	6000	-	-	-	6000
AD_PRES	-	-	-	24000	24000
SA_REP	-	26000	7000	-	33000

12.

```
SELECT
    d.department_name || ' - ' || l.city AS "Name-Location",
    COUNT(e.employee_id) AS "Number of People",
```

```
ROUND(AVG(e.salary), 2) AS Salary
FROM departments d
JOIN locations l ON d.location_id = l.location_id
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_name, l.city;
```

Output:

Name-Location	Number of People	SALARY
Sales - Mumbai	1	7000